ISE 218: Fundamentals of Information Technology

Practice Problems for Unit 6: Input/Output Systems

Directions: Below is a set of exercises and problems you should do for practice. Answering this set of questions should be just one part of your preparation for quizzes and exams. In addition to solving these problems you should:

- Review your homework assignments.
- Review the lecture notes and read the relevant sections of the course textbook.
- Do additional practice problems from the textbook; see the problems at the end of each chapter.
- Form a study group and test each other on the material.

1. Calculate the overall speedup of a system that spends 65% of its time on I/O with a disk upgrade that provides for 50% greater throughput.

$$S = \frac{1}{(1-f)+f/k} = \frac{1}{0.35+0.65/1.5} = 1.28 = 128\%$$

Overall speedup is 28%.

2. Suppose your company has decided that it needs to make certain busy servers 300% faster. Processes in the workload spend 80% of their time using the CPU and 20% on I/O. In order to achieve an overall system speedup of 300%, how much faster does the CPU need to be? Your final answer should be a whole number.

$$S = \frac{1}{(1-f)+f/k}$$

$$4 = \frac{1}{(1-0.8)+0.8/k}$$

$$4 = \frac{1}{0.2+0.8/k}$$

$$0.2+0.8/k = \frac{1}{4}$$

$$0.8/k = 0.25 - 0.2$$

$$0.8/k = 0.05$$

k = 0.8/0.05 = 16 times as fast (1500% faster)

3. Suppose your company has decided that it needs to make certain busy servers 400% faster. Processes in the workload spend 90% of their time using the CPU and 10% on I/O. In order to achieve an overall system speedup of 400%, how much faster does the CPU need to be? Your final answer should be a whole number.

$$S = \frac{1}{(1-f) + f/k}$$

$$5 = \frac{1}{(1-0.9) + 0.9/k}$$

$$5 = \frac{1}{0.1 + 0.9/k}$$

$$0.1 + 0.9/k = \frac{1}{5}$$

$$0.9/k = 0.2 - 0.1$$

$$0.9/k = 0.1$$

k = 0.9/0.1 = 9 times as fast (800% faster)

4. Using Amdahl's Law, show which is better: making 20% of the instructions in a program 80% faster, or making 80% of the instructions 20% faster.

   Making 20% of instructions 80% faster:

   $$S = \frac{1}{(1-0.2) + 0.2/1.8} = 1.098$$

   Making 80% of instructions 20% faster:

   $$S = \frac{1}{(1-0.8) + 0.8/1.2} = 1.153$$

   Making 80% of instructions 20% faster is the better option.

5. Suppose the daytime processing load of some computing server consists of 70% CPU activity and 30% disk activity. Customers are complaining that the system is slow. After doing some research, you have learned that you can upgrade your disks for $2,500 to make them 3 times as fast as they are currently. You have also learned that you can upgrade your CPU to make it 1.5 times as fast as it currently is for $3,000.

   a. Which would you choose to yield the best performance improvement for the least amount of money? Justify your answer through appropriate application of Amdahl's Law and a cost-benefit analysis similar to the one we did in lecture.

      $S_{CPU} = \frac{1}{(1-f)+f/k} = \frac{1}{(1-0.7)+0.7/1.5} \approx 1.3043$ or 30.43%

      $S_{Disk} = \frac{1}{(1-f)+f/k} = \frac{1}{(1-0.3)+0.3/3} \approx 1.25$ or 25%

      Choose the CPU upgrade:

      CPU = $3000/30.43% = $98.59 per 1% increase in performance Disk = $2500/25% = $100 per 1% increase in performance

   b. What is the break-even point for the upgrades? That is, what price would we need to charge for the CPU (or the disk – change only one) so the result was the same cost per 1% increase for both?

We want the price per 1% to be the same. A correct response must include *at least one* of the following analyses:

- If we change the CPU price, we have $X/30.43 = 100$, and $X = \$3043$. This means if the CPU costs $3043, we break even on cost per 1% improvement.
- If we change the price of the disk, we have $Y/25 = 98.59$, and $Y = 2464.75$. This means if the disk costs $2464.75, we break even on cost per 1% improvement.

6. Suppose a disk drive has the following characteristics:

- 8 surfaces
- 100,000 tracks per surface
- 1500 sectors per track
- 1024 bytes/sector
- Track-to-track seek time of 6 milliseconds • Rotational speed of 10,000 RPM.

a. What is the capacity of the drive in GB? Use base-2 units of measure, i.e., 1 GB = 2^{30} bytes.

8 surfaces × 100,000 tracks / surface × 1500 sectors / track × 1024 bytes/ sector = 1,228,800,000,000 bytes or 1144.4 GB

b. What is the access time in ms?

Access time = seek time + rotational delay

$$\text{Rotational delay} = \frac{\frac{60 \text{ seconds}}{\text{disk rotation speed}} \times \frac{1000 \text{ ms}}{\text{sec.}}}{2}$$

$$= \frac{\frac{60 \text{ seconds}}{10000} \times \frac{1000 \text{ ms}}{\text{sec.}}}{2}$$

$$= \frac{60/10}{2} = 3$$

Access time = 8.5 ms + 3 ms = 11.5 ms